

Is Earth Rare?

How we turned NASA's catalogue of 762 TESS exoplanets into a story about the worlds our telescopes simply cannot see.

Is Earth rare?

762 exoplanets confirmed by NASA's TESS mission. Only one dot is Earth's twin. Or is it even there at all?

TESS — the Transiting Exoplanet Survey Satellite — has been scanning the sky since 2018, finding planets that briefly dim their host star as they pass in front of it. But the telescope can only see what its design allows. This visualization explores what TESS finds, what it misses, and where Earth would sit if we could see everything.

Begin ↓

course **COM-480 Data Visualization** · EPFL · Spring 2026

team **Mohamed Bouchnak** 327170 **Hugo Heinkélé** 326309 **Hong Wei** 422972 **Gustavo Maia** 357154

site com-480-data-visualization.github.io/CosmoViz

repo github.com/com-480-data-visualization/CosmoViz

Is Earth rare, or does it just **look** that way?

That one question is the spine of the whole project. We had NASA's list of **762 exoplanets** confirmed by the TESS mission, and we wanted to show people how unlike Earth almost every single one of them is.

But the deeper we dug, the more we noticed something uncomfortable: a big part of the answer isn't about the planets at all. It's about us. TESS is great at spotting large planets on short, fast orbits around bright stars. A planet like Earth barely registers. So the website ended up telling two stories at once: *what the data shows*, and *what the telescope is blind to*.

Who we built it for. Not astronomers. We aimed at curious people and fellow students, so we leaned on plain explanations and visuals instead of dropping a giant scientific table on them, while trying hard not to bend the data, even when a tidier story was tempting.

The data. Everything comes from the NASA Exoplanet Archive (TESS confirmed planets, default parameters): each row is a planet with its radius, mass, temperature, orbital period and insolation, plus its host star. All the cleaning and exploration happens in `analysis.ipynb`.

762

CONFIRMED TESS
PLANETS

~620

WITH FULL
MEASUREMENTS

1

WITH A YEAR LONGER
THAN 300 DAYS

0.5%

CHANCE TESS WOULD
EVEN SEE EARTH

“The empty corner of our charts isn't missing data. **It's the shape of our blind spot.**”

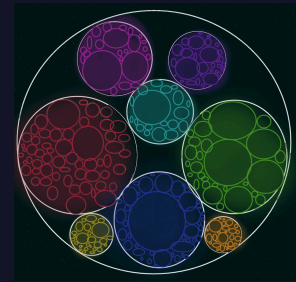
Our first idea was a toy. The data turned it into an argument.

In Milestone 1 the plan was a **planet browser**: Earth in the middle, surrounded by little white dots. Click a dot and you'd get a spider chart and a habitability score. We even sketched some “if-we-have-time” ideas, like clustering and a build-your-own-planet tool.

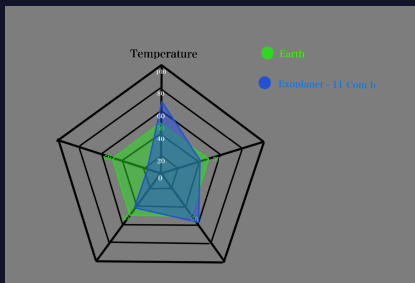
We kept almost all of these sketches. But once we asked what each one actually *told* the user, most of them quietly changed jobs.



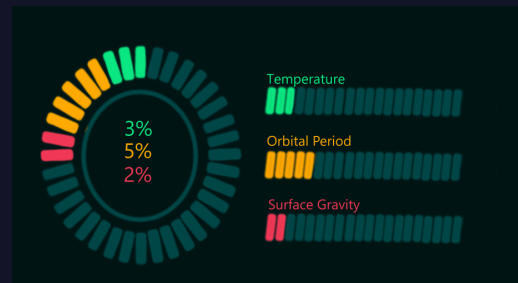
M1 ·the “universe of dots” concept



M1 ·clustering the planet types



M1 ·the spider / radar comparison



M1 ·the habitability score

Looking at the four sketches above, only one of them (the radar) survived more or less intact. The rest pointed us somewhere we didn't expect when we started. That turn is the subject of the next page.

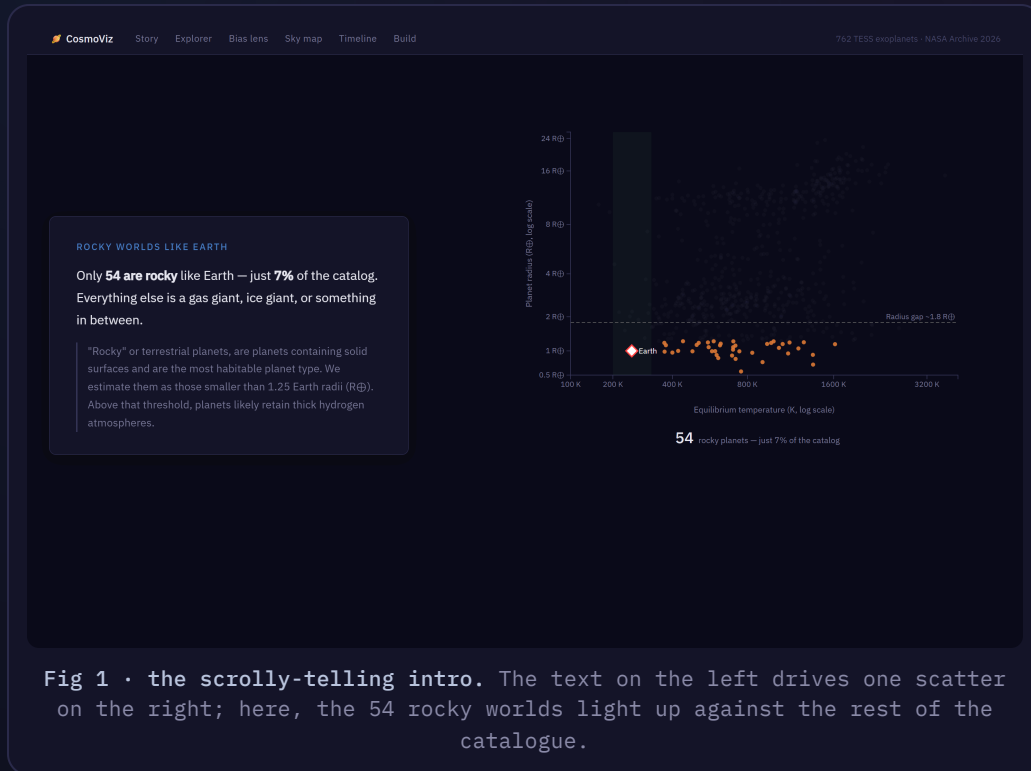
What changed, and **why**

- **The “universe of dots” split into two real charts.** The original dots were placed at random: pretty, but meaningless. So we made position carry data: the **sky map** puts each planet where it really sits in the sky, and the **Explorer scatter** places it by temperature and radius. Now *where* a dot is actually means something.
- **Clustering became a colour, not a chart.** A separate cluster plot just repeated what we already knew, so the planet’s size class (● Rocky → ● Gas Giant) turned into one consistent colour we reuse *everywhere*: legend, scatter, timeline, story. The colour links the views instead of adding another one.
- **The spider chart and score stayed, but grew up.** The radar moved from centre-stage into the detail card and the sandbox, because one radar can’t tell the *population* story. The score, which we’d basically made up, was rebuilt as a proper **Earth Similarity Index** (a weighted geometric mean, weights after Schulze-Makuch et al. 2011), so Earth lands on exactly 100 and we can defend the formula.
- **And the big one: we gave the project a thesis.** M1 was something to click around in. The final version makes a point. We added a guided **scrolly-telling** intro and rebuilt everything around *detection bias*, following the “martini-glass” idea from the storytelling lecture: lead with the takeaway, then hand over the controls.

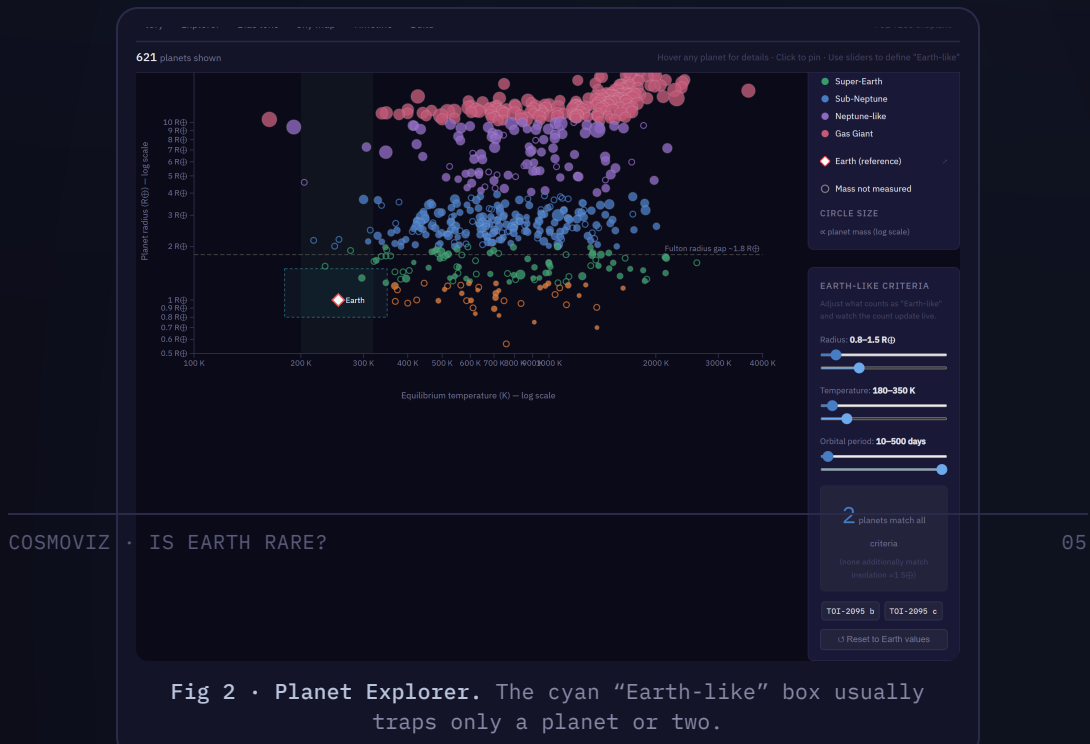
From a planet browser to a single argument: **the planets we can’t find are the whole point.**

First we guide you, then we let you loose

The story section scrolls you through the catalogue one beat at a time (all the planets, then the handful of rocky ones, then the sub-Neptunes that are *everywhere*, and finally the eerily empty “Earth-like” zone), and the same scatter re-colours itself at every step. Only after the punchline do we hand over control.



Then: Planet Explorer. Temperature versus radius on log axes, dot size for mass, colour for class, with the habitable zone and the Fulton radius gap marked. The sliders let *you* decide what “Earth-like” means, then watch how few planets survive your definition.



The same planets, **seen three more ways**

The **detection-bias lens** re-plots everything as orbital period versus radius, with the TESS ~27-day limit and an “Earth-analog zone” drawn in. The box is almost empty, and that emptiness is the argument. A year slider replays how the catalogue filled up.

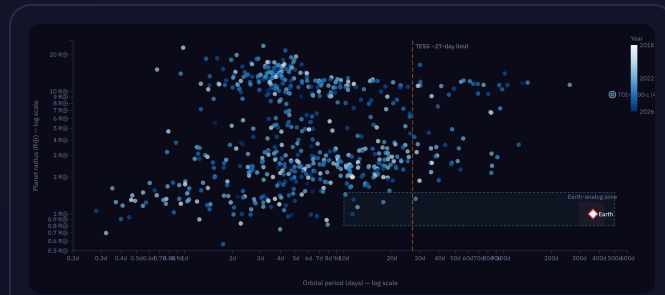


Fig 3 · **detection bias**. Hardly anything sits at Earth-like periods.

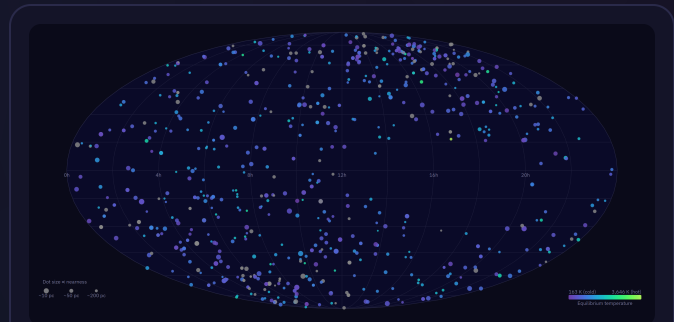


Fig 4 · **Mollweide sky map**. Real positions, not decoration.

The **sky map** places all 762 planets at their true coordinates, coloured by temperature and sized by distance, and you can even make out the pattern of where TESS looked. And the **sandbox** lets you build a world from scratch and watch its temperature, insolation, habitable zone and Earth-similarity update live; it makes painfully clear how easily habitability falls apart when you change the star.

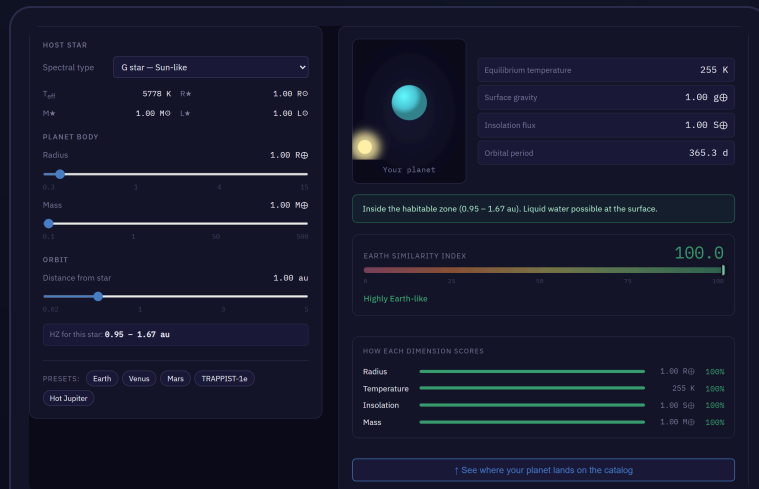


Fig 5 · **build your own planet**. Earth scores exactly 100; almost nothing else does.

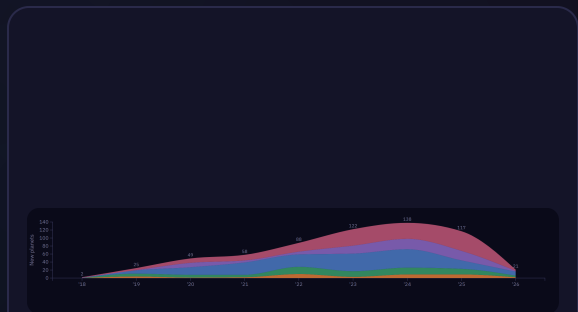


Fig 6 · **discovery timeline**. Click a band to filter every view.

How we designed it, and where it got hard

DESIGN CHOICES WE MADE

- **Position first.** Like the marks-and-channels lecture, the two most important numbers always go on position; mass (less reliable) goes on size, class on colour.
- **Log scales on purpose.** Radius, period and temperature span three to four orders of magnitude, so log/symlog axes are used and labelled, since a linear scale would crush every rocky planet into one corner.
- **One palette, and we checked it.** The five class colours were run through a colourblindness (deuteranopia) simulator; blue encodes discovery year, a cool ramp encodes temperature.
- **Earth is always the anchor.** The same white-diamond-with-red marker appears on every chart so you never lose your reference point.
- **Annotations do the talking.** The Fulton gap, the habitable zone and the TESS line are what turn a cloud of dots into a sentence.
- **Everything is linked.** A small event bus plus Crossfilter keep all six views in sync as you hover, select and filter.

WHAT FOUGHT BACK

- **Missing data.** Many planets lack a mass, temperature or insolation. We didn't delete or invent: all 762 stay on the sky map, the physical scatter uses the ~620 with full data, and "mass not measured" shows as a hollow circle. (*That's why some views say 762 and others ~620.*)
- **Telling a story about biased data.** The hardest part: showing that TESS *can't* find Earth analogues, without implying they don't exist.
- **Staying correct.** Temperature, insolation, the habitable zone and the ESI are derived from real physics (Stefan-Boltzmann, Kepler, Kopparapu) and checked against known planets.
- **Cosmic scale.** The numbers are huge and tiny at once, so log scales and an Earth-relative transform keep comparisons honest and readable.
- **Six live views at once.** ~760 animated dots across six linked views broke a lot early on; isolating each view behind the bus and using D3's enter/update/exit means one crash can't take down the rest.

Under the hood, and the team

The site runs on **D3.js v7** (scales, axes, shapes, stacks, zoom), **d3-geo-projection** for the Mollweide map, **Crossfilter** for the linked filtering and **Scrollama** for the scroll steps. It's plain ES modules with no build step, hosted as a static site on GitHub Pages, with the data prepared in a Jupyter notebook. The code is one module per view, all talking through a shared state/event bus, and each view fails on its own so it can't take down the others.

We split the work roughly evenly, about **25% each**. All four of us were in on the concept, the design reviews, this report and the screencast; the table below is just each person's main area.

MEMBER	SCIPER	MAIN AREA OF OWNERSHIP
Mohamed Bouchnak	327170	Data pipeline & EDA (<code>analysis.ipynb</code>): cleaning the NASA CSV, the missing-value strategy, the derived features (size class, stellar luminosity, habitable-zone ratio). Plus the project setup (module loader, shared state/event bus, the “Earth-like” criteria & sliders), and the README and repo.
Hugo Heinkélé	326309	The story spine: the scrolly-telling intro (Scrollama) and its scatter. The Planet Explorer view (temperature × radius), the live “Earth-like” box, and the hover/highlight linking that ties the views together.
Hong Wei	422972	The detection-bias view (period × radius) with the TESS limit and year animation; the Mollweide sky map (d3-geo-projection, temperature/distance encoding, system highlighting); and the discovery timeline with click-to-filter.
Gustavo Maia	357154	The habitability side: the Earth Similarity Index, the gauge and per-axis breakdown, and the radar chart. The “build your own planet” sandbox (temperature / insolation / habitable-zone physics). Plus the tooltips, detail card, and the visual design / CSS theme.